Week 4 Milestone Worksheet

Instructions: This worksheet will help you complete your Week 4 Milestone slide deck, so complete it before you start putting your deck together. Each part of this worksheet must correspond to the content that you will place into your finished slide deck.

In order to complete the assignment, you must follow these steps:

1. Complete this worksheet with the content that you will put on each slide.
2. Create a deck on Google Slides or Canva.
3. Copy and paste content from this worksheet into your slide deck, creating slides in line with the instructions shared in the Milestone.
4. Add any missing visuals to your slides in line with the instructions in the Milestone.
5. Make sure your slide deck is aesthetically pleasing and well-designed.
6. Get the share link to your slide deck, and add it here at the end of this worksheet. Make sure that the link is viewable by everyone.
7. Save, and submit this worksheet on Savanna.

SECTION A: SLIDE DECK CONTENT

Slide 1: Your Problem Statement

State your problem in 1 sentence. Your problem should be a sub-component of a GCGO. (As stated above, we recommend you use the same problem/GCGO as last,

but you can change it if you wish to re-do the steps in Week 3 Milestone.) For example, if your GCGO was climate change your problem might be "Plastic pollution in the city of Nairobi is growing." If your GCGO was healthcare then your problem might be "There is a huge shortage of physicians in Namibia," and if your GCGO was arts, culture, and design your problem might be "Visual artists across the African continent lack a platform to sell their work." (We also recommend that you use the same problem that you scoped and researched last week, but you can change it if you wish.) Then briefly describe WHY your problem matters in 1-2 sentences. Challenge yourself to state your WHY as clearly and succinctly as possible.

Slide 1 Title

Enhancing access to quality education through technology-driven learning solutions

Slide 1 Content

Slide 1: Problem Statement

Title: Bridging the Education Gap with Technology

Problem Statement: Many students in underserved communities lack access to quality education due to limited resources and outdated teaching methods.

Why This Matters:

Education is key to economic growth and social mobility.

Lack of access to quality education perpetuates cycles of poverty and inequality.

Technology-driven learning solutions can help bridge this gap and improve learning outcomes.

Slide 2: Your Personal Mission Statement

Make a personal connection with the problem, by sharing your personal mission statement, and why/how it connects with the chosen GCGO or problem stated in slide 1.

Slide 2 Title

My mission is to leverage technology to create inclusive and accessible learning opportunities for all students, regardless of their background or location.

Slide 2 Content

My mission is to create equal learning opportunities for all by leveraging technology to enhance access to quality education. I am passionate about bridging educational gaps, empowering students with innovative digital tools, and ensuring that every learner, regardless of their background, has the

resources they need to succeed. By integrating technology-driven solutions, I aim to foster a future where education is inclusive, engaging, and accessible to everyone

Slide 3: Your Target Community

Make it real. Describe 1 person or 1 specific group of people impacted by this problem and how it impacts them.

Slide 3 Title

The target community includes underserved students in low-income areas and educators in schools with limited resources, where technology can bridge the educational gap

Slide 3 Content

The target community consists of underserved students in rural and low-income urban areas, as well as educators in schools with limited access to modern educational resources, where technology can provide equal learning opportunities and enhance educational outcomes.

Why This Community?

These groups face significant educational disparities, and technology can help bridge the gap, providing equal learning opportunities for all.

Slide 4: Your Research

Describe how you've been researching the problem to learn more about it. You should do so by sharing your research questions/objectives and a brief description of your research plan. Provide at least 2 research objectives and research questions each that are related to your research into the GCGO or problem of choice.

Slide 4 Title

Research and Insights

Slide 4 Content

Research Questions:

1. **What are the main barriers to education in rural and low-income urban areas?**
2. **How can technology be integrated effectively in classrooms with limited resources to improve student learning outcomes?**

Research Plan:

- **Conduct secondary research through studies, reports, and articles on educational disparities in underserved areas.**
- **Analyze case studies of successful technology integration in resource-constrained schools.**
- **Survey teachers and students to understand the specific challenges and potential solutions for using technology in education.**

Slides 5-7: Facts About Your Problem

On each of these slides, present 1 piece of factual information that your web research revealed about your problem. If you are presenting numerical data, consider including a graph or other data visualization. Cite the (reliable) source as a footnote at the bottom of each page. You should select information to share that is compelling, relevant, and accurate.

Slide 5 Title

Facts About the Problem (Ethiopia)

Slide 5 Content

Teacher Shortages and Overcrowded Classrooms

📌 **Fact:** Ethiopia has a **student-to-teacher ratio of 60:1 in many rural areas**, significantly higher than the recommended **40:1** by UNESCO.

📊 **Visualization Idea:** A bar graph comparing Ethiopia's student-to-teacher ratio with other African countries.

🔹 **Impact:** Overcrowded classrooms reduce individual attention for students, leading to lower learning outcomes and higher dropout rates.

📖 **Source:** World Bank Education Report, 2023

Slide 6 Title

Technology and Infrastructure Challenges

Slide 6 Content

📌 **Fact:** Only **12% of students in rural Ethiopia have access to computers**, and fewer than **10% have internet access** in schools.

📊 **Visualization Idea:** A pie chart showing the percentage of students with and without access to digital learning tools.

🔹 **Impact:** The lack of technology prevents students from gaining essential digital skills, further widening the education gap.

📖 **Source:** Ethiopian Ministry of Education Report, 2023.

Slide 7 Title

Limited Access to Quality Education in Ethiopia

Slide 7 Content

Fact: Over **25% of Ethiopia's rural population lacks access to quality education,** with fewer than **40% of children in rural areas completing secondary school.**

📊 **Visualization Idea:** A bar chart comparing rural vs. urban school completion rates in Ethiopia.

🔹 **Impact:** Limited access to education hinders economic mobility and job opportunities for young people in rural areas.

📖 **Source:** UNESCO Institute for Statistics, 2023.

Slide 8: Framework Applied to Break the Problem

Using either the Cool Hands Warm Heart or the McKinsey Problem-Solving Framework, break the problem down into smaller components. Describe the framework used and the components created.

Slide 8 Title

Breaking Down the Problem: Applying the McKinsey Problem-Solving Framework

Slide 8 Content

- **Framework Used:** McKinsey Problem-Solving Framework
- **Components of the Problem:**
    1. **Infrastructure:** Lack of schools, classrooms, and learning materials in rural areas.
    2. **Teacher Training:** Insufficient training and support for educators, leading to poor

Slides 9-11: Your Solutions

For each of these slides, share 1 possible way you could address your problem (so that you share a total of 3 potential solutions across 3 slides). Each idea may be obvious, extremely bold and creative, or anywhere in between. At least 2 of your potential solutions must use technology. Be sure that each of your 3 ways is "SMART"- Specific, Measurable, Achievable, Relevant, and Time-Bound.

Slide 9 Title

Digital Learning Platform

Slide 9 Content

Develop and deploy a mobile digital learning platform that provides offline-accessible, curriculum-aligned educational content for rural students in Ethiopia.

- **Specific: Create an app offering interactive lessons, quizzes, and multimedia content based on Ethiopia's national curriculum, designed for low-bandwidth environments.**
- **Measurable: Target to reach 10,000 rural students within 18 months, tracking engagement through user metrics and completion rates.**
- **Achievable: Partner with local NGOs, telecom providers, and educational experts to curate and distribute content.**
- **Relevant: Directly addresses the technology gap and limited access to quality educational resources in underserved areas.**

- **Time-Bound: Complete platform development and pilot testing within 12 months, with a full rollout by 18 months.**

Slide 10 Title

Teacher e-Learning and Training Program

Slide 10 Content

Launch an e-learning-based professional development program to train rural teachers on integrating digital tools into their classrooms.

- **Specific:** Design an online training course featuring webinars, interactive modules, and best practices for using technology in teaching.
- **Measurable:** Aim to train 500 rural educators in the first year, with improvements measured via pre- and post-training assessments and classroom observation scores.
- **Achievable:** Collaborate with local education authorities and universities, leveraging existing e-learning platforms and expertise.
- **Relevant:** Equips teachers with necessary digital skills to enhance the learning experience, mitigating the teacher shortage and quality issues.

Slide 11 Title

Community Learning Centers

Slide 11 Content

Establish community learning centers that serve as hybrid hubs for both digital and traditional educational resources.

- **Specific:** Set up 50 centers in rural regions equipped with computer labs, libraries, and spaces for in-person tutoring and online classes.
- **Measurable:** Each center will serve at least 200 students, with the goal of increasing literacy and numeracy scores by 15% over two years.
- **Achievable:** Utilize existing community spaces and secure funding through government grants and partnerships with local and international educational foundations.
- **Relevant:** Provides a tangible, local resource that supplements under-resourced schools and bridges both technological and infrastructural gaps.
- **Time-Bound:** Launch the first 10 centers within 6 months and complete all 50 centers within 24 months.

Slide 12: Your User Story

For 1 of your 3 solutions, write a simple and clear user story that describes your problem and include acceptance criteria for each story. Use plain English that is easy for your target audience to understand. For example, for a challenge in the climate change space, you could have the following as your user story:

- As a conscious consumer, when I shop online, I want to easily identify and choose Eco-friendly products to reduce my carbon footprint.

Acceptance criteria for this user story could be:

- Users should be able to filter and sort products based on sustainability labels, certifications, or environmental impact indicators.
- Each product listing should display relevant information about its environmental impact, such as carbon emissions, water usage, or recyclability.

Make your user story clear, concise, testable, and measurable. Your acceptance criteria should be “SMART”—Specific, Measurable, Achievable, Relevant, and Time-Bound—and should focus on what’s important to the end user.

Slide 12 Title

User Story – Digital Learning Platform

Slide 12 Content

As a rural student in Ethiopia, when I use the digital learning platform, I want to access interactive, curriculum-aligned lessons offline so that I can learn effectively despite unreliable internet connectivity.

Acceptance Criteria:

- **Offline Accessibility:** The platform must cache educational content on the device, ensuring that 100% of the core lessons are accessible without an internet connection.
- **Curriculum Alignment:** At least 90% of the available lessons and quizzes should align with Ethiopia's national curriculum, as verified by local educational experts.
- **Interactive Engagement:** Each lesson must include a minimum of 2 interactive quizzes with instant feedback, and average user engagement per lesson should be at least 20 minutes.
- **Progress Tracking:** The platform should display a progress dashboard that tracks quiz scores and lesson completion rates for each user.
- **Timely Implementation:** The pilot version of the platform should be deployed in 3 rural schools and reach at least 10,000 students within 12 months.

Slide 13: Your Solution KPIs

Describe what exactly you would measure in order to see if one of your solutions was working, and how you would go about measuring it.

Slide 13 Title

Key Performance Indicators (KPIs) for the Digital Learning Platform

Slide 13 Content

What We Will Measure & How:

Adoption Rate:

Metric: Number of active users and downloads from target rural communities.

Measurement: Track app downloads and monthly active users using in-app analytics.

Target: Reach 10,000 active users within 12 months.

Engagement Rate:

Metric: Average session duration and lesson completion rate per user.

Measurement: Analyze user activity logs to monitor the time spent per session and the percentage of lessons completed.

Target: Achieve an average session duration of 20 minutes and an 80% lesson completion rate.

Offline Accessibility Utilization:

Metric: Percentage of core lessons accessed offline.

Measurement: Use analytics to track the ratio of offline content access compared to total content usage in low-connectivity areas.

Target: Ensure 90% of the core curriculum lessons are accessed

User Satisfaction:

Metric: User feedback and satisfaction ratings.

Measurement: Conduct periodic surveys with students and teachers, collecting ratings on a scale of 1 to 5.

Target: Maintain an average satisfaction rating of 4 or higher.

Academic Improvement:

Metric: Change in student performance (e.g., test scores) before and after using the platform.

Measurement: Collaborate with local schools to compare standardized test scores from pilot groups at the start and after one academic year.

Target: Achieve at least a 15% improvement in test scores within one academic year.

Each of these KPIs is SMART—Specific, Measurable, Achievable, Relevant, and Time-Bound—ensuring we can accurately assess the impact and success of our digital learning solution.offline by users in targeted regions.

Slide 14: Your Mission and Your Solution

Return to your WHY. Restate your WHY and share why one of your solutions could make an important difference.

Slide 14 Title

My Mission and Our Digital Learning Platform

Slide 14 Content

Revisiting Our WHY:
My mission is to leverage technology to create equal learning opportunities for every student, especially those in underserved communities, so that they can break free from cycles of poverty and contribute to sustainable development.

Why Our Digital Learning Platform Matters:

Empowering Students: By providing curriculum-aligned, interactive lessons accessible offline, our solution directly addresses the educational challenges faced by students like Amina in rural Ethiopia.

Bridging Gaps: It removes barriers related to limited resources and unreliable internet access, ensuring that every student can engage with quality educational content.

Measurable Impact: With clear KPIs such as increased user engagement, improved test scores, and higher satisfaction ratings, we can track meaningful progress in academic performance and digital literacy.

This solution is not just a tool—it's a step towards transforming lives and building a more equitable future in education.

Slide 15: Appendix

Appendix. Share at least 2 pieces of information from your web research. This can be any information that you feel is important/relevant but that you didn't get to share in the main body of your presentation. For example, you may choose to include additional data or facts about your problem or solution(s), information about related solutions shown to be successful, identify unknowns and/or unanswered questions, list organizations that are doing similar work, etc. Please make sure that you list all sources for the information that you share on this slide.

Slide 15 Title

Appendix – Additional Research & Insights

Slide 15 Content

Connectivity Challenge in Rural Ethiopia:
Research indicates that less than 7% of households in rural Ethiopia have reliable internet access, underscoring the necessity for offline-capable digital learning solutions.
Source: UNESCO Institute for Statistics, 2023[1]

Impact of Digital Interventions:
Studies from similar contexts reveal that technology-integrated educational programs can boost student performance by 15-20% over one academic year, provided there is adequate teacher training and community support.
Source: International Development Research Centre (IDRC), 2022[2]

UNESCO Institute for Statistics, 2023.

[2] International Development Research Centre (IDRC) Report, 2022

Slide 16: Appendix 2

Appendix 2: List the info from Steps 1 & 2 (Target Demographic and User Persona) above in the same slide.

Slide 16 Title

Appendix 2 – Target Demographic & User Persona

Slide 16 Content

Target Demographic:

Underserved Students: Primarily students in rural and low-income urban areas of Ethiopia who lack access to quality, technology-enhanced education.

Educators: Teachers in these regions who work with limited resources and outdated educational tools.

User Persona:

Name: Amina

Age: 14

Location: A small rural village in Ethiopia

Background & Challenges:

Amina is an ambitious student with dreams of becoming a doctor.

She faces significant educational barriers due to outdated textbooks, a lack of digital tools, and insufficient teaching resources.

These challenges limit her learning opportunities and hinder her ability to

achieve her academic and career aspirations.

STOP

Please use all this content to create your slide deck. Add any missing visual elements, and fix the aesthetic and design of the slide deck.

SECTION B: SLIDE DECK LINK

Slide Deck Link

Once your slide deck is complete, ensure the following:

- The content you placed here in the worksheet is the same as that in your deck.
- Your deck meets all the requirements stated in the Milestone instructions, including the required visuals.
- Make sure the slides are visually appealing and well-designed. See the rubric for criteria related to design.

Paste the Slide Deck Link Below

Once you have completed this worksheet:

1. Save to .pdf.
1. Rename it per the instructions.
2. Upload to Savanna as your Week 4 Milestone Submission.
3. **Celebrate a job well done!**